# Lec 6

Tabular Method

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$$= \boxed{-5e^{-4} + 1}$$

**Example 4 - Tabular Method**: In Example 2 we have to apply the Integration by Parts Formula multiple is a convenient way to "book-keep" our work. This is done by creating a table. Let's see how by examinin again.

Evaluate

$$\int x^2 e^x \, dx.$$

Let $f(x) = x^2$ and $g(x) = e^x$. Then,

| Differentiate $f(x)$ | | Integrate $g(x)$ |
|---|---|---|
| $x^2$ | $+$ | $e^x$ |
| $2x$ | $-$ | $e^x$ |
| $2$ | $+$ | $e^x$ |
| $0$ | | $e^x$ |

Then the integral is,

$$\int x^2 e^x \, dx = +x^2 \cdot e^x - 2x \cdot e^x + 2 \cdot e^x + C$$

**Example 5**: Find the integral

$$\frac{1}{\pi}\int_0^{\pi} x^3 \cos(nx)\; dx,$$

where $n$ is a positive integer.

Let $f(x) = x^2$ and $g(x) = \cos(nx)$. Then,

| Differentiate $f(x)$ | | Integrate $g(x)$ |
|---|---|---|
| $x^3$ | $+$ | $\cos(nx)$ |
| $3x^2$ | $-$ | $\frac{1}{n}\sin(nx)$ |
| $6x$ | $+$ | $-\frac{1}{n^2}\cos(nx)$ |
| $6$ | $-$ | $-\frac{1}{n^3}\sin(nx)$ |
| $0$ | | $\frac{1}{n^4}\cos(nx)$ |

Then the integral is,

$$\begin{aligned}
\frac{1}{\pi}\int x^3\cos(nx)\,dx &= \frac{1}{\pi}\left[+x^3\cdot\frac{1}{n}\sin(nx) - 3x^2\cdot\left(-\frac{1}{n^2}\right)\cos(nx) + 6x\cdot\left(-\frac{1}{n^3}\right)\sin(nx) - 6\cdot\frac{1}{n^4}\cos(nx)\right]\Bigg|_0^{\pi} \\
&= \frac{1}{\pi}\left[\frac{x^3}{n}\sin(nx) + \frac{3x^2}{n^2}\cos(nx) - \frac{6x}{n^3}\sin(nx) - \frac{6}{n^4}\cos(nx)\right]\Bigg|_0^{\pi} \\
&= \frac{1}{\pi}\left[\left(0 + \frac{3\pi^2}{n^2}\cos(n\pi) - 0 + \frac{6}{n^4}\right) - \left(0 + 0 - 0 - \frac{6}{n^4}\right)\right] \\
&= \frac{1}{\pi}\left[\frac{3\pi^2(-1)^n}{n^2} - \frac{6(-1)^n}{n^4} + \frac{6}{n^4}\right] \\
&= \boxed{\frac{3}{\pi}\,\frac{\pi^2 n^2(-1)^n - 2(-1)^n + 2}{n^4}}
\end{aligned}$$

**Example 6 - Recurring Integrals**: Find the integral

$$\int e^x \sin(x)\,dx.$$

We need to apply Integration by Parts twice before we see something:

(1)

$$u = e^x \qquad dv = \sin(x)$$
$$du = e^x\,dx \qquad v = -\cos(x)$$

(2)

$$u = e^x \qquad dv = \cos(x)$$
$$du = e^x\,dx \qquad v = \sin(x)$$

$$\begin{aligned}\int e^x \sin(x)\,dx &= -e^x\cos(x) + \int e^x\cos(x)\,dx \\ &= -e^x\cos(x) + \left(e^x\sin(x) - \int e^x\sin(x)\,dx\right) \\ &= -e^x\cos(x) + e^x\sin(x) - \int e^x\sin(x)\,dx\end{aligned}$$

Notice that now the integral we are interested in, $\int e^x\sin(x)\,dx$, appears on both the left and right hand side of the equation. So, if we add this integral to both sides we get

$$\Longrightarrow \quad 2\int e^x\sin(x)\,dx = e^x(-\cos(x) + \sin(x))$$

$$\Longrightarrow \quad \int e^x\sin(x)\,dx = \boxed{\frac{e^x(\sin(x)-\cos(x))}{2}}$$